

Sentiment Analysis

Week 12 – Programming

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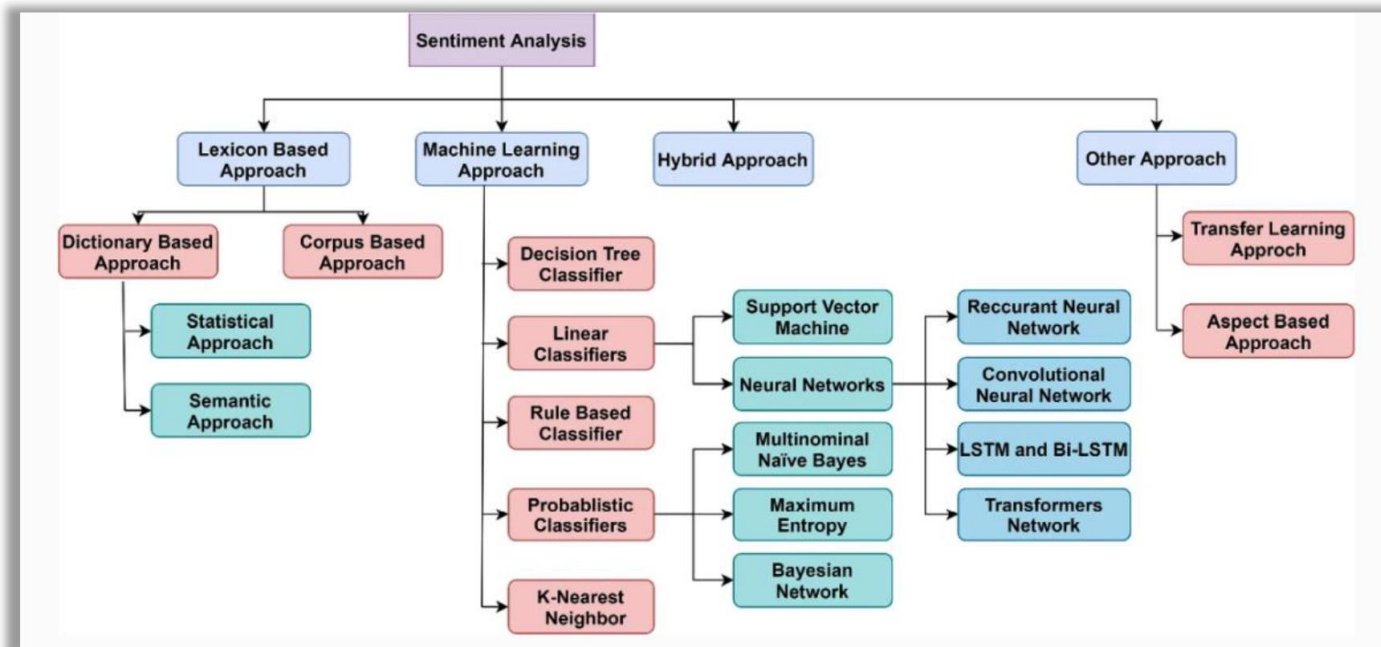
4. Final examination

❖ Week 12 Objective

- Sentiment Analysis
 - Lexicon-based
 - Machine learning-based through training
 - BERT-based Sentiment Analysis

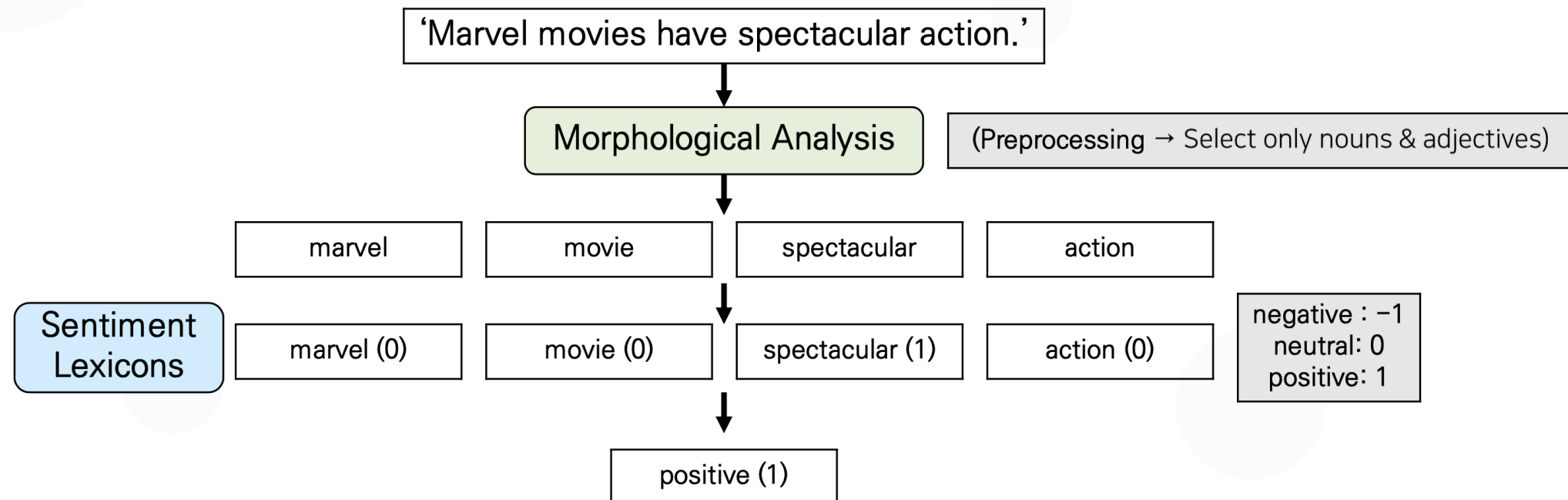
❖ Understanding Sentiment Analysis

- Sentiment analysis refers to the process of analyzing subjective information such as opinions, evaluations, and attitudes expressed in text.
- Sentiment is a subjective emotion towards a particular subject, usually classified into three categories: positive, neutral, and negative.
- There are various methodologies for sentiment analysis, which can generally be categorized into lexicon-based analysis and machine learning-based analysis.



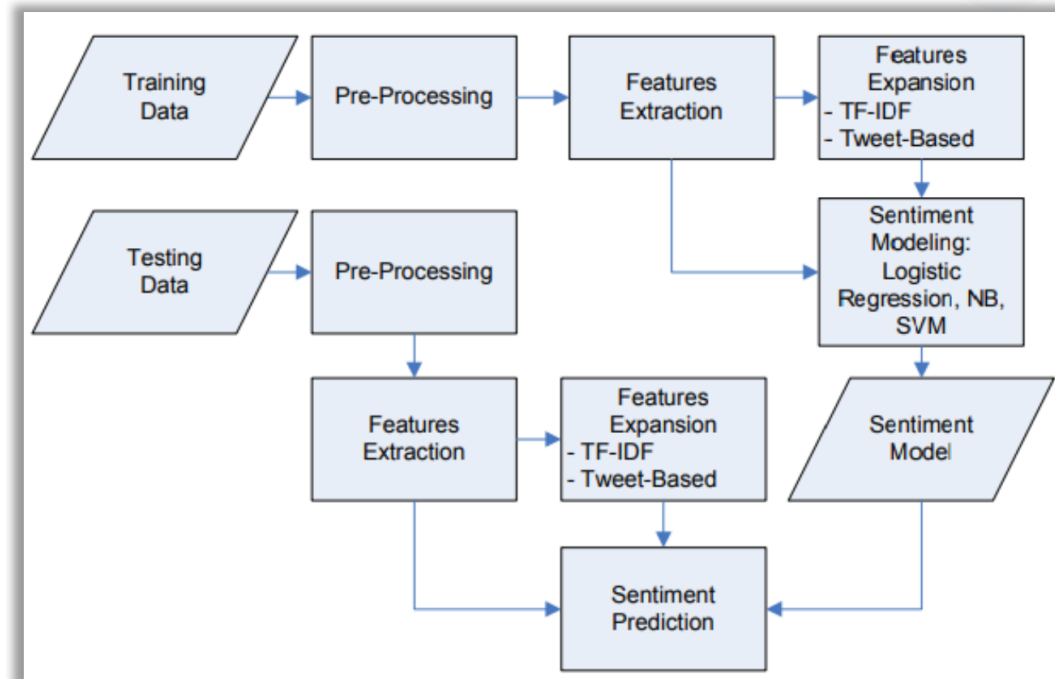
❖ Lexicon-based Sentiment Analysis

- This method involves creating a sentiment lexicon by assigning positive or negative sentiments to words like nouns, adjectives, and verbs. Sentiment analysis is then performed using this lexicon.
- While lexicons can be built manually, external sentiment lexicons are often used due to the time and effort required for manual creation.



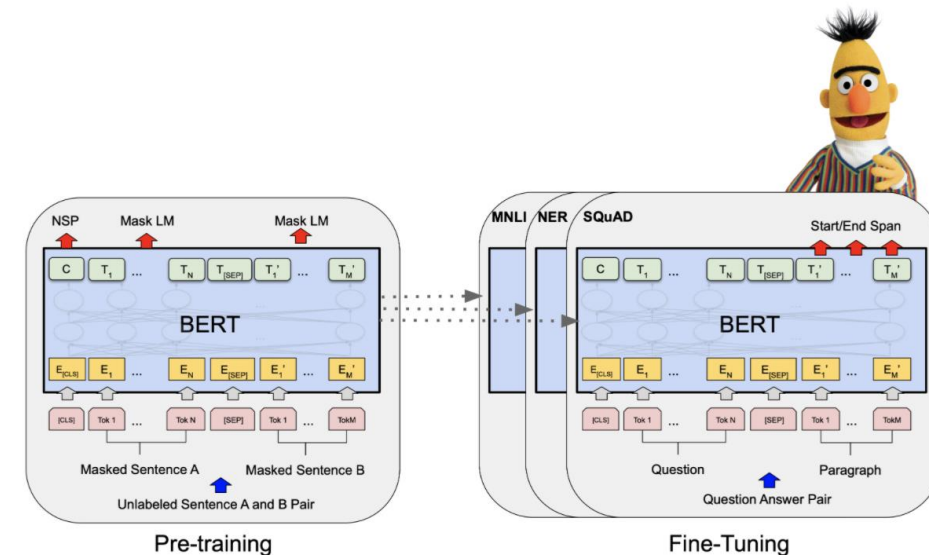
❖ Sentiment Analysis Based on Machine Learning through Training

- To perform sentiment analysis using machine learning, a labeled training dataset is needed. For example, if analyzing movie reviews, the dataset should pair each review with a sentiment label.
- This is a document classification problem with classes like 'positive', 'negative', and 'neutral'.
- With the dataset, various classification algorithms can be used to train a model to predict sentiment.



❖ BERT-based Sentiment Analysis

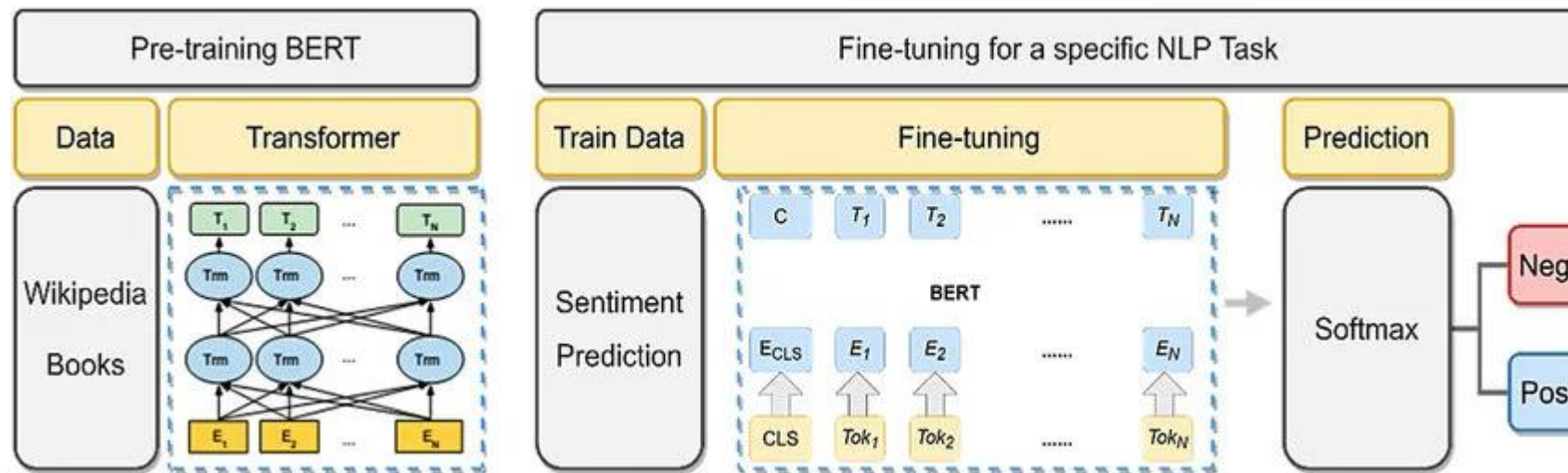
- What is BERT?
 - Bidirectional Encoder Representations from Transformers (Google, 2018)
 - Pre-training + Fine-tuning approach
- Key Advantage
 - Lexicon-based: Fixed dictionary
 - Machine Learning: Manual features
 - BERT: Contextual understanding



Input	[CLS]	my	dog	is	cute	[SEP]	he	likes	play	#ing	[SEP]
Token Embeddings	E _[CLS]	E _{my}	E _{dog}	E _{is}	E _{cute}	E _[SEP]	E _{he}	E _{likes}	E _{play}	E _{#ing}	E _[SEP]
Segment Embeddings	E _A	E _A	E _A	E _A	E _A	E _A	E _B	E _B	E _B	E _B	E _B
Position Embeddings	E ₀	E ₁	E ₂	E ₃	E ₄	E ₅	E ₆	E ₇	E ₈	E ₉	E ₁₀

❖ BERT Sentiment Analysis Pipeline

- Three Steps
 1. Tokenization: [CLS] text [SEP]
 2. Fine-tuning: Pre-trained → Task-specific
 3. Classification: Softmax → Sentiment scores

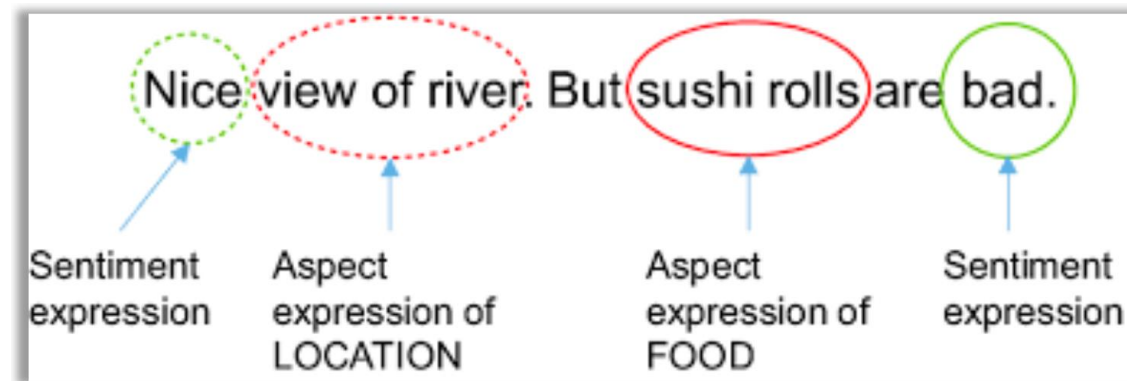


❖ Programming

- Link: https://drive.google.com/drive/folders/1CQscMDXjxiGNYx-9P3cyn-RAF_DMnCMY?usp=drive_link

❖ Advanced: Aspect-Based Sentiment Analysis (ABSA)

- Definition: Classifying sentiment towards specific aspects (topics) in a text.
- Importance: A single text may contain multiple aspects, each requiring independent sentiment analysis.
- Challenge: Accurately aligning sentiment expressions with their corresponding aspects for effective opinion mining.



❖ Advanced: Sentiment Analysis using Pre-trained Language Models

- Reference link: https://medium.com/@manjindersingh_10145/sentiment-analysis-with-bert-using-huggingface-88e99deec9a

❖ Assignment 6: Sentiment Analysis

- Colab Link:
https://drive.google.com/file/d/1IU8NSaAPBqAudpkVUcW0iby65JDy4aAG/view?usp=drive_link
- Save the file as 'Assignment_YourName_YourStudentID.ipynb' and submit it to the 'Programming Assignment 6 – Sentiment Analysis' section under Assignments
- Deadline: 23:59 on Tuesday, November 25th

❖ Exam Materials Guide

- Code and materials corresponding to the exam scope will be organized and uploaded
- An announcement will be posted once the materials are available on the LMS. Please check back regularly for updates.

Q & A

Thank you for your attention. Any questions are welcome!

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